Robustness of the Cell2Sentence Model: Evaluating the Role of Rank Order under Poisson Noise and Gene Sequence Shuffling.

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1 Abstract

The Cell2Sentence framework encodes single-cell RNA sequencing data as ordered sequences of gene names ranked by expression magnitude, enabling pretrained language models to perform cell type classification. A theoretical advantage of this rank-based representation is its inherent robustness to technical noise, since moderate perturbations to expression magnitudes need not alter the relative gene ordering. This work evaluates that robustness empirically through two complementary experiments. First, Poisson noise is injected into raw count matrices at varying fractions of expressed genes (0%–100%), simulating realistic scRNA-seq variability. Second, gene rank order within cell sentences is directly disrupted through intrablock shuffling at varying block sizes, isolating the contribution of rank structure. Results show that the C2S-Scale-Gemma-2-2B model maintains high accuracy (82.4%–86.4%) under all levels of Poisson noise, consistent with rank order remaining largely intact under this perturbation. In contrast, intrablock shuffling causes a clear monotonic accuracy decline from 86.4% to 56.4% as rank information is progressively destroyed. Together, these findings provide empirical support for the view that the robustness of Cell2Sentence to expression-level noise stems from its rank-based representation.

2 Introduction

The rapid advancement of single-cell RNA sequencing (scRNA-seq) technologies has generated an unprecedented volume of high-dimensional transcriptomic data. A central task in the analysis of such data is cell type classification, the automated assignment of biological identity to individual cells based on their gene expression profiles. Traditionally, this problem has been approached through clustering algorithms, dimensionality reduction techniques, and supervised classifiers trained on curated marker gene sets [6]. While these methods have proven effective in

controlled settings, they are often sensitive to technical artifacts inherent to scRNA-seq data, including dropout events, sequencing depth variability, and amplification noise, all of which can corrupt the absolute expression values on which many methods rely [10].

In parallel, the field of natural language processing has witnessed transformative advances driven by large-scale pretrained language models (LLMs). Their success in capturing complex sequential dependencies and adaptability across diverse domains has motivated researchers to explore the application of these models in the field of bioinformatics [8]. Cell2Sentence is one such approach: it encodes each cell's expression profile as an ordered sequence of gene names ranked by expression magnitude, referred to as a *cell sentence*, and fine-tunes causal language models on these sequences for downstream biological tasks [5]. A theoretical advantage of this rank-based representation is its inherent robustness to noise because it preserves relative gene expression order rather than absolute values, moderate perturbations to expression magnitudes need not alter the resulting sequence significantly [5]. To assess robustness to realistic technical variability, Poisson noise is introduced into the raw count data and its effect on cell type prediction accuracy is evaluated. Gene rank order is then systematically altered through intrablock shuffling to isolate the contribution of rank structure. It is hypothesized that if robustness is mediated by rank, then (a) Poisson noise will not substantially reduce accuracy because rank is preserved, and (b) direct disruption of rank will reduce accuracy proportionally to the degree of shuffling.

3 Related work

3.1 Single-Cell RNA Sequencing and Technical Noise

Single-cell RNA sequencing enables the measurement of genome-wide transcript counts at cellular resolution. A fundamental downstream task is the annotation of individual cells with biologically meaningful type labels. Early approaches relied on manual marker gene inspection, but the scale of modern datasets has driven the development of automated classification methods, including graph-based clustering with marker gene scoring [9], machine learning classifiers such as support vector machines and random forests [1], and reference-based label transfer methods that map query cells onto annotated cell datasets [4]. A persistent challenge across all these approaches is the susceptibility to technical noise. In particular, scRNA-seq count data is characterized by overdispersion, dropout (zero-inflation), and batch effects, all of which can introduce systematic and stochastic errors into expression measurements [6]. Analysis of nine public datasets indicated that, after adjusting for differences in capture and sequencing efficiency between cells, technical noise in single-cell UMI counts can be reasonably modeled using a Poisson distribution [7].

3.2 Cell2Sentence

Cell2Sentence is a framework that adapts pretrained LLMs to single-cell transcriptomic data by converting numerical gene expression profiles into sequences of gene names. For each cell, genes are ranked according to their expression magnitude and represented as an ordered sequence, thereby preserving relative expression patterns while making the data compatible to be utilized by standard LLM architectures. This transformation allows existing causal language models to be fine-tuned on cell sentences, leveraging natural language pretraining to improve performance on tasks relevant to single-cell biology. Fine-tuned models can accurately predict cell-type labels from cell sentences. Another advantage is that generated cell sentences can also be mapped back into the gene expression space via a simple inverse transformation. Cell2Sentence thus provides a flexible, scalable approach to combine natural language modeling with transcriptomic data, broadening the applicability of LLMs in computational biology [5].

4 Approach and Experiments

4.1 Overview

This work evaluates the robustness of the Cell2Sentence framework under two complementary perturbations: Poisson gene expression noise and targeted disruption of within-cell gene rank via intrablock shuffling. While Poisson noise reflects realistic technical variability and affects both gene identity and rank order, intrablock shuffling isolates the impact of rank disruption without altering gene composition. A pretrained Cell2Sentence foundation model is used for cell type prediction. The full pipeline spans four stages: data acquisition, preprocessing, noise or rank perturbation, and model evaluation.

4.2 Dataset Acquisition

Similar to the Cell2Sentence setup a subset of the cross-tissue immune cell dataset published by Domínguez Conde et al [3], which profiled immune cells across multiple human tissues using single-cell RNA sequencing, was used. From this dataset, cells from two donors (A29 and A31) were selected, yielding a total of 29,773 cells covering a diverse set of immune cell types. The dataset is provided in the AnnData (H5AD) format.

4.3 Preprocessing

Raw transcript count matrices were preprocessed exactly according to the Cell2Sentence framework. Standard single-cell RNA-seq quality control was applied using Scanpy: cells expressing fewer than 200 genes and genes detected in fewer than 3 cells were removed. The remaining counts were normalized by scaling each cell to a total count of 10,000, followed by a base-10 log trans-

formation. No deviations from the Cell2Sentence preprocessing protocol were introduced.

4.4 Perturbation Methods

4.4.1 Poisson Noise Injection

To simulate technical variability commonly observed in real-world scRNA-seq experiments, Poisson noise was injected into the raw count matrices prior to normalization. For each cell and for a given noise fraction $f \in \{0, 0.1, 0.2, \ldots, 1.0\}$, a proportion f of the non-zero genes was randomly selected, and their raw counts were replaced with Poisson-distributed samples drawn using the original counts as the rate parameter (λ). Following noise injection, the modified count matrices were preprocessed using the exact same Cell2Sentence preprocessing pipeline described above. This resulted in 11 datasets representing noise levels from 0% (reference condition, no added noise) to 100% in increments of 10%, enabling systematic evaluation of model robustness as a function of input corruption.

4.4.2 Intrablock Shuffling of Gene Rankings

As a second perturbation regime, the rank-order structure of cell sentences was directly disrupted through an intrablock shuffling procedure, without altering the underlying expression values. For each cell sentence, the ordered list of genes was divided into contiguous blocks of fixed size $b \in \{10, 25, 50, 100, 200, 300\}$. The genes within each block were subsequently randomly shuffled, thereby eliminating local rank information while preserving global set membership. This shuffling procedure was restricted to the top 300 genes, given that the model utilizes only the first 200 genes; extending the range to 300 accommodates genes that may drop in or out of the top-200. The remainder of the sequence was left unchanged.

4.5 Model Evaluation

4.5.1 Model

The pretrained `vandijklab/C2S-Scale-Gemma-2-2B` model, a 2-billion-parameter autoregressive language model based on the Gemma-2 architecture and trained by Van Dijk Lab for cell type prediction from Cell2Sentence representations, was used for all evaluations. The model was accessed via the `CSModel` wrapper provided in the `Cell2Sentence` library.

4.5.2 Cell Sentence Similarity Analysis

Prior to model evaluation, the effect of the perturbations on the cell sentence representation itself was quantified independently of model predictions. For each cell, the clean and perturbed top-200 gene lists were compared using two complementary metrics:

- **Jaccard Index:** Measures overlap in gene identity between clean and noisy gene sets, defined as

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

- **Spearman Rank Correlation:** Measures preservation of expression ranking among genes shared between the clean and noisy top-200 gene lists, quantifying how well relative expression order is maintained.

These metrics are computed across all 29,773 cells for each noise level and block size, and then averaged to obtain a mean value representing overall input degradation.

4.5.3 Evaluation Protocol

To control computational cost, 250 cells were uniformly sampled from each dataset (noise levels 0%–100% and block sizes 0, 10, 25, 50, 100, 200, 300) using a fixed random seed to ensure exact reproducibility and to guarantee that the same 250 cells were evaluated across all perturbation conditions. A single reference vocabulary, derived from the unperturbed (0%) dataset, was used for all `CSData` objects to ensure consistent tokenisation across conditions.

For each sampled cell, the model was prompted with its corresponding cell sentence, truncated to the top-200 genes to remain consistent with the original configuration. The model generated one cell type prediction per cell. Predicted labels were post-processed by removing the special token `<ctrl100>` and stripping trailing punctuation and whitespace prior to comparison with ground-truth annotations. Cell type prediction accuracy was computed as the fraction of cells for which the predicted label exactly matched the reference annotation. Accuracy is reported across all perturbation conditions.

5 Results and Analysis

5.1 Cell Sentence Similarity Analysis

To assess the impact of each perturbation on cell sentence representations, the Jaccard index and Spearman rank correlation were computed between the top-200 gene lists of clean and perturbed cells ($n = 29{,}773$) across both perturbation regimes.

5.1.1 Poisson Noise

Gene set overlap (Jaccard index). The results is table 1 show that gene set similarity decreased approximately linearly with increasing noise, from 1.000 at the baseline to 0.628 at 100% noise, representing a 37% reduction in overlap. This steady decline indicates a predictable, noise-dependent disruption of the gene set, with genes progressively dropping in and out of the top-200 as noise increases.

Table 1: Similarity for different noise percentages

Noise level (%)	Mean Jaccard	Mean Spearman
0	1.000	1.000
10	0.914	0.988
20	0.848	0.974
30	0.796	0.960
40	0.754	0.945
50	0.721	0.931
60	0.694	0.917
70	0.672	0.905
80	0.654	0.894
90	0.640	0.884
100	0.628	0.877

Rank preservation (Spearman correlation). In contrast, rank correlation remained highly robust. Spearman's ρ declined only modestly from 1.000 at baseline to 0.877 at 100% noise, a reduction of 12.3%. This indicates that although noise alters gene membership in the top-200 set, the relative ordering of highly expressed genes is largely preserved.

Implications for model robustness. These results highlight a key trade-off: while rank information remains stable under noise, gene set composition changes substantially. Consequently, downstream robustness will depend on whether classification relies more on gene identity or rank structure.

5.1.2 Intrablock Shuffling

Table 2: Similarity for different block sizes

Block size	Mean Jaccard	Mean Spearman
10	1.000	0.998
25	1.000	0.984
50	1.000	0.937
100	1.000	0.750
200	1.000	0.001
300	0.500	0.000

Gene set overlap (Jaccard index). For all block sizes up to 200, the Jaccard index remains exactly 1.000, confirming that intrablock shuffling does not alter gene set membership within the top-200; only the ordering is affected. At block size 300, the Jaccard index drops to 0.500 because shuffling within a block of 300

extends beyond the 200-gene evaluation window, causing genes from positions 201-300 to displace genes originally ranked within the top-200.

Rank preservation (Spearman correlation). In contrast to the stable gene set membership, rank correlation degrades sharply as block size increases. At block size 10, ρ remains near-perfect at 0.998, indicating that shuffling within very small windows barely disrupts the overall rank structure. As block size grows, rank correlation declines steeply: 0.984 at size 25, 0.937 at size 50, 0.750 at size 100, and effectively 0.001 and 0.000 at sizes 200 and 300, respectively, representing near-total destruction of rank information.

Comparing the two perturbation strategies, intrablock shuffling isolates the contribution of rank order by keeping gene identity intact, whereas Poisson noise jointly perturbs both gene identity and rank order.

5.2 Classification Accuracy

Cell type prediction accuracy was evaluated across all perturbation conditions on a fixed sample of 250 cells, using the `C2S-Scale-Gemma-2-2B` foundation model. Results for both the Poisson noise and intrablock shuffling experiments are shown in Figure 1. The exact numerical results are provided in Tables 3 and 4 in the Appendix.

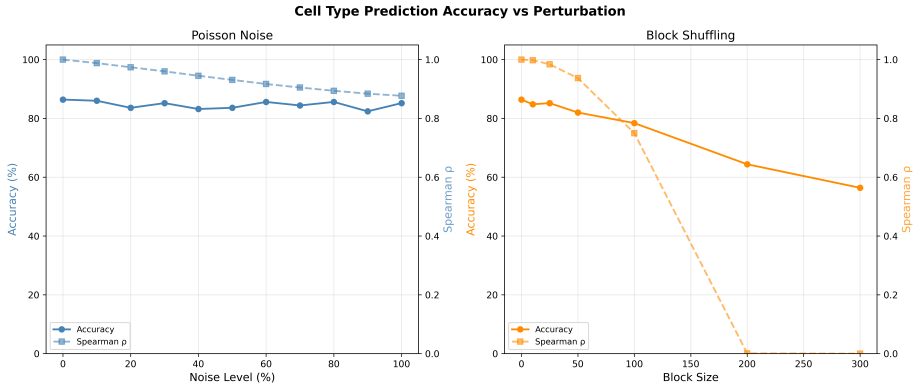

Figure 1: Cell type prediction accuracy as a function of perturbation strength. Left: Poisson noise injection at varying fractions of expressed genes. Right: Block shuffling at varying block sizes.

5.2.1 Poisson Noise

The model demonstrates strong and consistent robustness to Poisson noise across all tested levels. Baseline accuracy at 0% noise is 86.4% (216/250), which is consistent with the previously reported baseline performance [2]. Accuracy

remains within a narrow band of 82.4%–86.4% across all noise levels. The lowest observed accuracy is 82.4% at 90% noise, representing a drop of only 4 percentage points from baseline. Notably, there is no monotonic degradation with increasing noise; accuracy fluctuates without a consistent downward trend, suggesting that the perturbations introduced by Poisson noise are largely absorbed by the rank-based cell sentence representation. This is consistent with the similarity analysis, which showed that Spearman rank correlation remained above 0.877 even at 100% noise, indicating that the relative gene ordering is largely preserved under this perturbation.

5.2.2 Intrablock Shuffling

In contrast to the flat profile observed under Poisson noise, intrablock shuffling reveals a clear and monotonic degradation in accuracy as rank order is increasingly disrupted. At small block sizes (10 and 25), accuracy remains close to baseline at 84.8% and 85.2% respectively, consistent with the near-perfect Spearman correlations (0.998 and 0.984) observed at these levels. As block size increases, accuracy declines steadily: 82.0% at size 50, 78.4% at size 100, and dropping substantially to 64.4% at size 200 and 56.4% at size 300. Block size 300 no longer represents a purely rank-based perturbation, as the Jaccard index drops to 0.500 (Table 2), highlighting that the result reflects both rank disruption and gene set alteration and should be interpreted accordingly. The sharp decline between block sizes 100 and 200 mirrors the collapse in Spearman correlation from 0.750 to effectively 0.001 at those same sizes, demonstrating that prediction performance closely follows rank preservation. Crucially, since gene identity remains fixed throughout, this degradation is strongly consistent with rank order being the primary driver of prediction performance, providing direct evidence that gene ordering carries substantial predictive signal in the Cell2Sentence representation.

6 Conclusions

The Cell2Sentence framework proves robust to the form of technical noise most commonly encountered in scRNA-seq data. Under Poisson noise, the C2S-Scale-Gemma-2-2B model retained high accuracy (82.4%–86.4%) across all noise levels, consistent with rank order remaining largely intact despite progressive disruption of gene set composition. Intrablock shuffling, which isolates rank disruption by keeping gene identity fixed, caused a clear monotonic accuracy decline from 86.4% to 56.4%, with the steepest drop occurring where Spearman correlation collapsed between block sizes 100 and 200. Together, these results suggest that Cell2Sentence can be reliably applied to raw scRNA-seq data without dedicated noise correction, as its rank-based representation naturally absorbs realistic levels of technical variability.

The evaluation is limited to a single dataset from two donors and a single model architecture, meaning results may not generalize across datasets or models.

Furthermore, the 250 cells were sampled uniformly without stratification by cell type, meaning rare cell types may be underrepresented. Additionally, all accuracy estimates were obtained using a single fixed random seed, and no statistical tests were conducted. Given the small sample and single seed, differences fewer than 5 percentage points should not be over-interpreted. Future work should explore larger sample sizes and more diverse datasets, as well as robustness to other technical noise sources such as batch effects and dropout.

7 References

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8 Appendix

External Support: The code to calculate the Jaccard distance and Spearman correlation between cell sentences was obtained from Nils Rosenboom.

Usage of AI tools for writing or coding: Claude was used for coding assistance and writing guide paraphrasing sentences.

Table 3: Cell type prediction accuracy under Poisson noise

Noise level (%)	Accuracy	Correct/Total
0	0.864	216/250
10	0.860	215/250
20	0.836	209/250
30	0.852	213/250
40	0.832	208/250
50	0.836	209/250
60	0.856	214/250
70	0.844	211/250
80	0.856	214/250
90	0.824	206/250
100	0.852	213/250

Table 4: Cell type prediction accuracy under block shuffling

Block size	Accuracy	Correct/Total
0	0.864	216/250
10	0.848	212/250
25	0.852	213/250
50	0.820	205/250
100	0.784	196/250
200	0.644	161/250
300	0.564	141/250